Introduction to Data Lifecycle

In the modern digital world, **data is continuously created and used** in almost every activity.
From student attendance to online shopping and air quality monitoring, data passes through a **series of stages** before it becomes useful information. This complete journey is known as the **data lifecycle**.

Stages of Data Lifecycle

- Data Generation
- Data Collection
- Data Storage
- Data Processing
- Data Analysis
- Visualization & Decision-Making

Each stage is important and depends on the previous one.

What is Data?

- Data refers to **raw facts and figures** that are collected but do not have meaning until they are processed and analyzed.
- **Forms of Data**

Numerical: 45, 78, 120

Text: Delhi, Good, Excellent

Date/Time: 01-01-2026

Boolean: Yes/No, True/False

- **Practical Example**
- In an Excel sheet: Student Name, Marks, Date of Exam
- At this stage, these values are **only data**, not conclusions.

Data Generation

- **Data generation** refers to the process by which data is created as a result of activities, observations, measurements, or interactions.
- In today's digital world, data is generated **continuously** sometimes without us even realizing it. Whenever we use a device, fill a form, make a payment, or record information, data is produced.
- Data can be generated in two main ways:
- **Automatically** (by machines, sensors, and software)
- **Manually** (by humans entering or recording information)

Some common sources of Data Generation

1. **Sensors**

Sensors are electronic devices that detect physical or environmental changes and convert them into data. These devices work continuously and generate large volumes of data without human intervention.

Common Types of Sensors:

- **Air quality sensors**
- **Temperature sensors**
- **Humidity sensors**
- **Motion sensors**

Example: sensors installed in cities measure pollution levels such as PM2.5, PM10, carbon monoxide, etc. These sensors generate pollution values **every hour**.

2. Online Transactions

- **Online transactions** generate data whenever money, goods, or services are exchanged digitally. Each action performed on an app or website leaves behind a digital record like Paying college fees online, recharging mobile phones or ordering food using apps.

Whether it is Shopping apps (Amazon, Flipkart, Myntra) and Banking systems (UPI, net banking, debit/credit cards) every **UPI payment** generates transaction data such as:

- Transaction amount
- Date and time
- Sender and receiver details
- Transaction ID

This data helps banks in tracking payments, preventing fraud, analyzing spending habits.

3. Educational Systems

Educational systems generate data through academic and administrative activities in schools and colleges. This data helps institutions monitor performance and make decisions.

Types of Educational Data:

- Attendance records
- Marks and grades
- Assignment submissions

Example:

When teachers mark attendance or enter exam marks, **student academic data** is created daily.

This data helps track student performance, Used to prepare report cards.

Note: Data is often generated **automatically** (sensors, apps) or **manually** (attendance, marks entry).

4. Surveys and Feedback

Surveys and feedback systems are used to collect opinions, views, and experiences from people. This type of data is usually qualitative in nature.

Tools Used:

- Google Forms
- Online questionnaires
- Feedback portals

Example:

Student feedback forms collected after exams or workshops generate **opinion data** such as: Satisfaction level, Suggestions for improvement and teaching quality feedback.

- This data helps organizations improve quality and make better decisions.

5. Social media platforms

Social media platforms and online activities are one of the biggest sources of data generation today. Every action such as like, comment, share, search, post performed by a user knowingly or unknowingly creates data that is stored and analyzed.

Data such as time spent on content, likes, shares, and comments, search history and preferences is used for various purpose.

- Platforms recommend videos and posts based on user interest
- Companies use this data for targeted advertisements
- Helps understand trends and popular topics among students

Data Collection

Data collection is the process of **gathering generated data from multiple sources into one place**.

- **Example**

A university collects:

- Attendance from teachers
- Marks from departments
- Feedback from students

All data is merged into one system.

Methods of data collection

- **1. Manual Collection**
- Paper forms
- Registers
- Interviews
- Surveys
- Sheets

Example:

Teacher enters attendance into Excel.

2. Online Forms

- **Online forms** are digital tools used to collect data from people through the internet. They allow users to enter information easily using a computer or mobile phone. The data collected through online forms is stored electronically, making it easy to organize, analyze, and share.
- Online forms are widely used because they save **time, paper, and effort** and can collect responses from many people at the same time.
- Most Common tool: Google Forms

- **Steps to Collect Data Using Google Forms**
- **Create the Form**
  - Open Google Forms
  - Add a title and description
  - Insert questions (short answer, MCQs, ratings, etc.)
- **Share the Form**
  - Generate a link and Share it
- **Collect Responses**
  - Fill and submit the form
  - Data is collected in real time
- **Automatic Data Storage**
  - Responses are stored automatically
  - Linked directly to **Google Sheets**
- **Analyze the Data**
  - Now you can view charts and summaries
  - Data can be downloaded for further analysis

3. Automated Systems

- **Automated systems** are systems that collect data automatically using machines, cameras, or smart devices, without requiring humans to manually enter the data. These systems continuously observe activities and record data in real time, making data collection **fast, accurate, and reliable**.
- In automated data collection, once the system is set up, data is gathered **continuously and automatically** as events occur.
- IoT devices
- CCTV systems

Example:
Traffic cameras collect vehicle count data.

Using automated systems **large volumes of real-time data** can be collected with high accuracy and minimal human effort, making them a major source of data in modern digital environments.

4. APIs

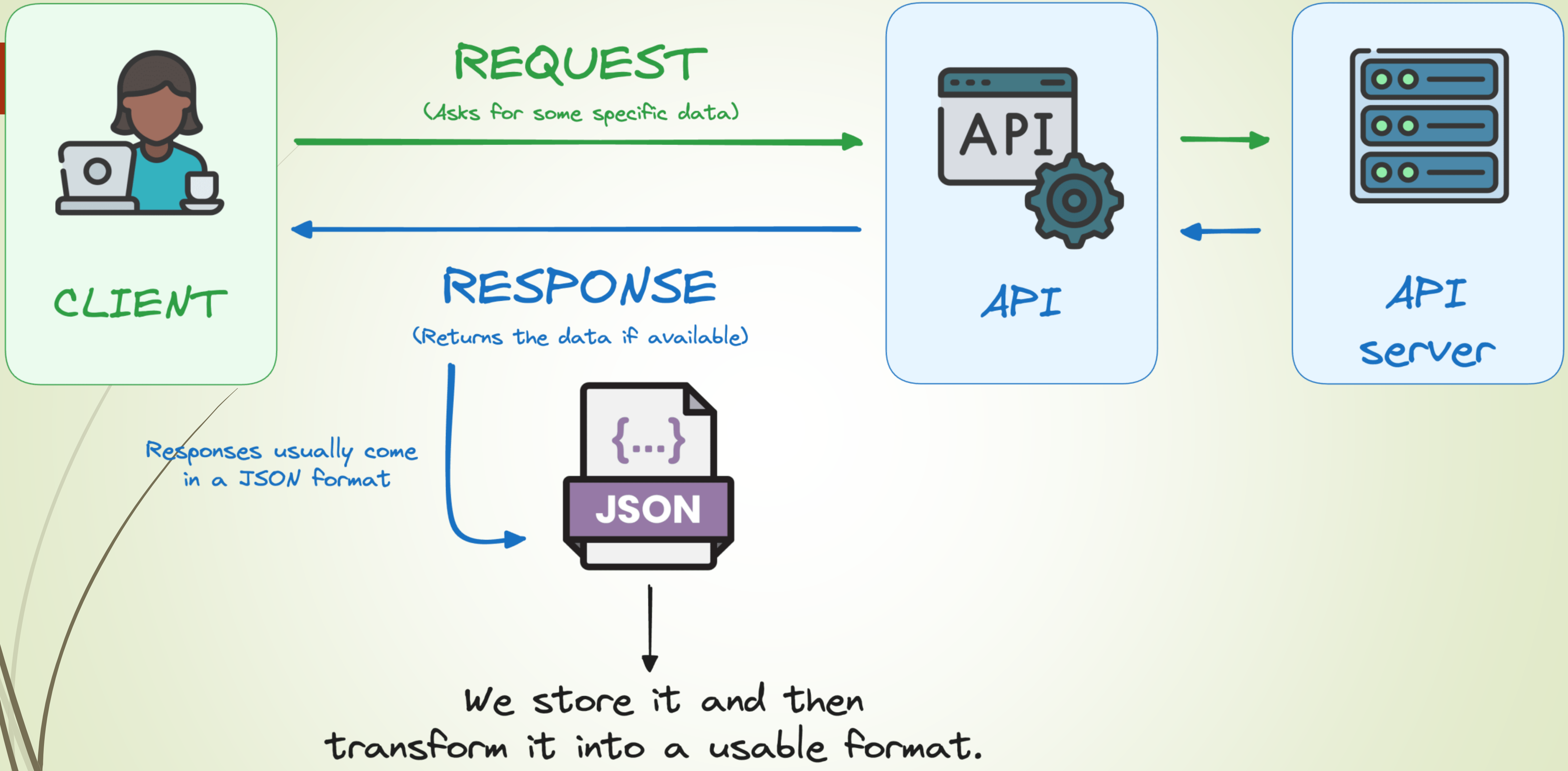
- **Application Programming Interface (API)** is a set of rules that allows one software application to communicate with another and exchange data automatically. APIs are widely used to collect data from online platforms, databases, and services without manual effort. Using APIs, data can be collected **directly from the source**, ensuring accuracy and real-time updates.
- APIs play a crucial role in data collection by acting as **digital bridges** that enable different software systems to communicate, exchange, and retrieve data automatically and in a standardized way. This eliminates manual data gathering, providing access to vast amounts of real-time, structured information
- **Example:**
  Weather apps collect data from meteorological APIs.

API:

https://jsonplaceholder.typicode.com/posts

https://wttr.in/Delhi?format=j1

https://stooq.pl/q/l/?s=aapl.us&f=sd2t2ohlcv&h&e=json


REQUEST
(Asks for some specific data)
CLIENT
RESPONSE
(Returns the data if available)
API
API
API server
Responses usually come in a JSON format
{...}
JSON
We store it and then transform it into a usable format.

Steps to Fetch Data Through API in Excel

- Excel can fetch data from APIs using its **Get Data / Power Query** feature. Most APIs return data in **JSON format**, which Excel can easily read.

Step 1: Get the API URL

- Choose a working API (example: weather, AQI, stock data)
- Copy the **API endpoint (URL)**

Step 2: Open Excel

Open **Microsoft Excel**

Open a **new blank workbook**

Step 3: Go to Get Data Option

- Click on the **Data** tab
- Select **Get Data**
- Choose **From Other Sources**
- Click **From Web**

This option is used to connect Excel with APIs and web services.

Step 4: Paste the API URL

- Paste the copied API URL into the dialog box
- Click **OK**
- Excel now sends a request to the API.

Step 5: Load the JSON Data

- Excel will detect that the data is in **JSON format**
- A **Navigator / Power Query window** opens
- Click on **Record** or **List** to expand the data

This step converts raw API data into readable tables.

- **Step 6: Transform the Data (Optional but Important)**
- In Power Query Editor:
- Expand columns, Rename column names and Remove unnecessary fields

This helps clean and organize the data before loading it.

- **Step 7: Load Data into Excel Sheet**
- Click **Close & Load**
- Data is loaded into an Excel worksheet as a **table**
- Now the API data is available inside Excel!
- **Step 8: Refresh Data (Very Important)**
- Right-click anywhere in the table
- Click **Refresh**
- Excel fetches **latest data from the API automatically**.

Key Roles of APIs in Data Collection

- **Automation:** Automatically fetches data, saving time and effort.
- **Structured Data Delivery:** Delivers specific information in formats like JSON, making it easy to parse.
- **Controlled Access:** Ensures secure, authorized access to data without exposing internal systems.
- **Integration:** Connects different apps (e.g., pulling weather or social media data) to create a unified view.
- **Real-time Updates:** Keeps information up-to-date instantly rather than relying on manual, periodic updates.

Storage

- **What is Data Storage?**

Data storage means **saving data in a place so that it can be used later**.

Just like:

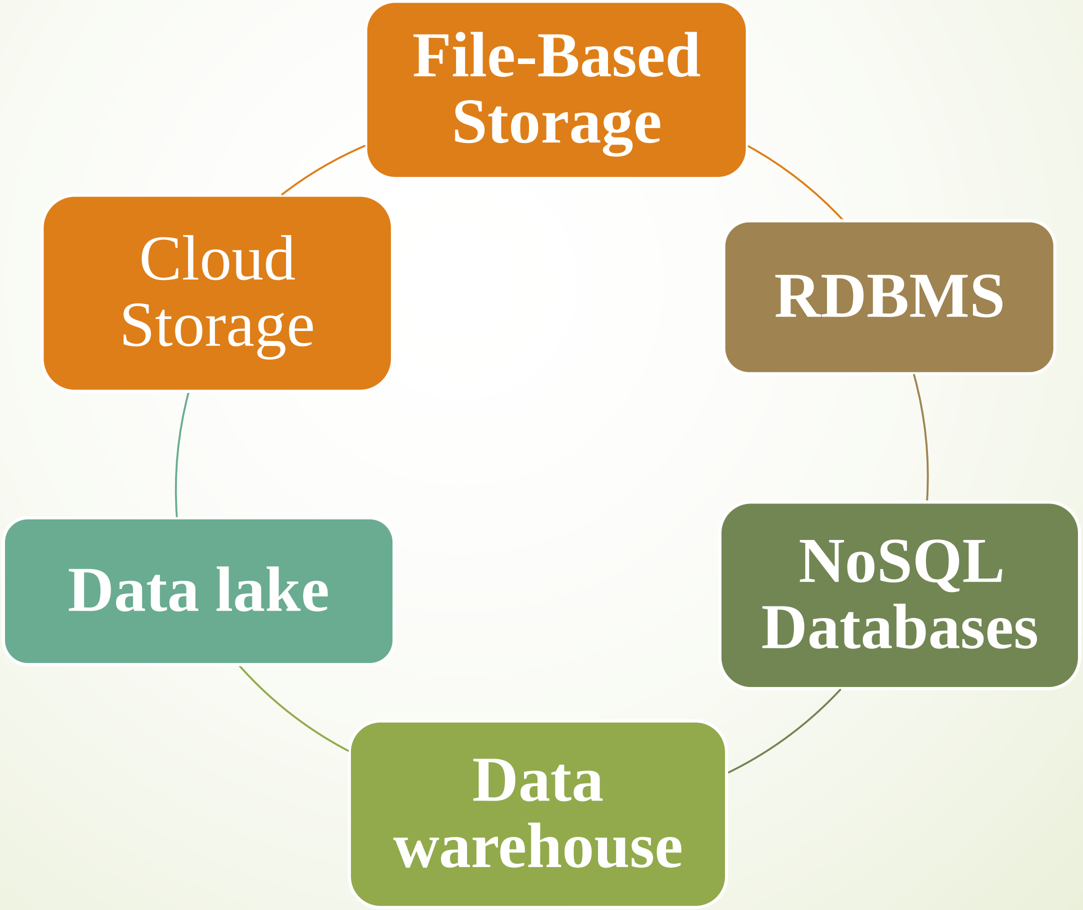
- We store books in cupboards
- We store clothes in wardrobes
- **Computers store data in storage systems**
- Student marks must be stored so teachers can generate results later.

Why Do We Need Data Storage?

- To **save information permanently**
- To **access data anytime**
- To **analyze data later**
- To **protect data from loss**
- To **share data with others**

Types of Data Storage

Data storage can be broadly classified into different types based on how and where data is saved and how it is accessed. Each type of data storage serves a specific purpose depending on the nature of the data and the requirements of the system.



File-Based Storage

File-based storage is one of the simplest and most commonly used types of data storage. In this method, data is stored in the form of files such as

- Data is stored in **files** like:
- Excel (.xlsx)
- CSV (.csv)
- Text (.txt)

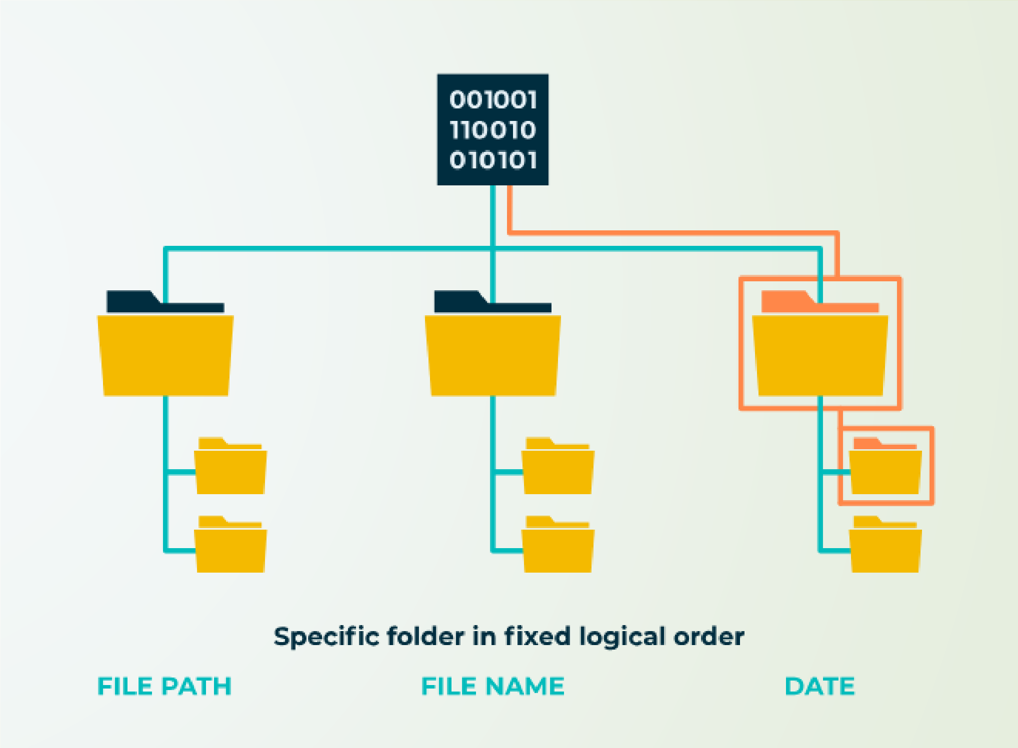
Each file is saved in a folder on a computer or storage device.

This type of storage is easy to use and understand,

especially for beginners.

Example:

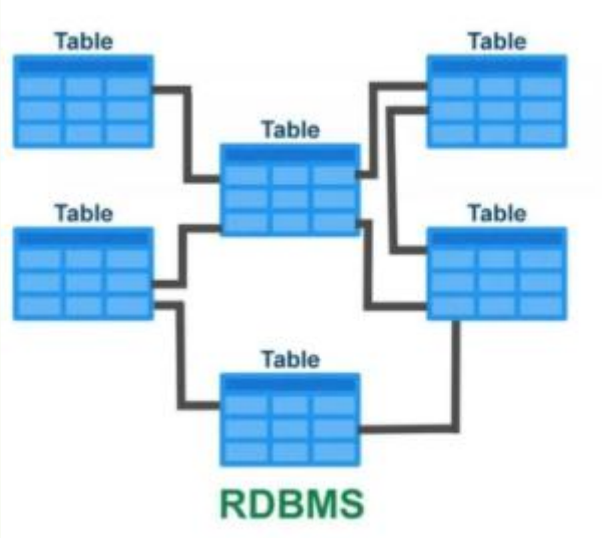
- Student data in Excel sheet
- Attendance in CSV file



Relational Database Management System (RDBMS)

RDBMS is a type of database where data is stored in an organized and structured manner using tables. Each table consists of rows and columns, similar to an Excel sheet. Every row represents a single record, and every column represents an attribute of that record. Because of this fixed structure, data becomes easy to store, search, update, and manage.

Some common examples of relational databases are:



- MySQL
- Oracle
- SQL Server
- PostgreSQL

These databases use SQL (Structured Query Language)

to create tables, insert data, and retrieve information.

Relational databases ensure data consistency, accuracy, and security, and they support powerful query operations using SQL. They are best suited for structured data where the format remains consistent.

NoSQL Databases

NoSQL database storage is designed to handle unstructured and semi-structured data that does not fit well into traditional table formats. Instead of rows and columns, data is stored in formats such as key-value pairs, documents, or graphs.

Examples of NoSQL databases include:

- MongoDB
- Cassandra
- Firebase

This type of storage is highly flexible and scalable, making it suitable for applications that deal with large volumes of data such as social media platforms, real-time analytics systems, and mobile applications.

NoSQL storage allows data to be stored without a fixed schema, which makes it easier to handle changing data structures.

Data warehouse

- Data warehouse storage is used to store large amounts of historical and processed data that is mainly used for analysis and reporting. Unlike regular databases that handle daily transactions, data warehouses are optimized for querying and analyzing data.
- They collect data from multiple sources such as databases, files, and applications, and store it in a centralized system.
- Organizations use data warehouses to analyze business trends, generate reports, and support decision-making.
- Examples include Amazon Redshift, Google BigQuery, and Snowflake.

Need for Data Warehousing

- **Handling Large Data Volumes**: Traditional databases store limited data (MBs to GBs), while data warehouses are built to handle huge datasets (up to TBs), making it easier to store and analyze long-term historical data.
- **Enhanced Analytics**: Databases handle transactions; data warehouses are optimized for complex analysis and historical insights.
- **Centralized Data Storage**: A data warehouse combines data from multiple sources, giving a single, unified view for better decision-making.
- **Trend Analysis**: By storing historical data, a data warehouse allows businesses to analyze trends over time, enabling them to make strategic decisions based on past performance and predict future outcomes.
- **Business Intelligence Support:** Data warehouses work with BI tools to give quick access to insights, helping in data-driven decisions and improving efficiency.

Database vs Datawarehouse

Aspect	Database	Data Warehouse
Purpose	Primarily for transactional operations	Primarily for analytical operations
Data Type	Handles structured data	Handles structured and unstructured data
Schema	Generally, follows a normalized schema	Often follows a denormalized schema
Data Volume	Usually handles smaller data volumes	Handles large volumes of historical data
Performance	Optimized for read and write operations	Optimized for complex queries and reporting
Data Freshness	Emphasizes real-time data	Emphasizes historical and periodic data
Query Complexity	Supports simpler, real-time queries	Supports complex, analytical queries
Data Transformations	Limited emphasis on data transformations	Involves significant data transformations
Usage	Used for day-to-day operations	Used for decision-making and analysis
Data Model	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)

Data lake

- Data lake storage is a type of storage that holds raw data in its original format until it is needed.
- This data can be structured, semi-structured, or unstructured, such as text files, images, videos, audio files, and sensor data.
- Data lakes are commonly used in big data and machine learning applications where data is analyzed later.
- Unlike data warehouses, data lakes do not require data to be cleaned or structured before storage. This makes data lakes flexible but also requires careful management to avoid data confusion.

DATA LAKE vs DATA WAREHOUSE

DATA LAKE

Data	Users	Use cases
unstructured	Data Scientists, Data Analysts	Stream Processing, Machine Learning, Real time analysis

DATA WAREHOUSE

Data	Users	Use cases
Structured	Business Analysts	Batch Processing, BI, Reporting

Raw

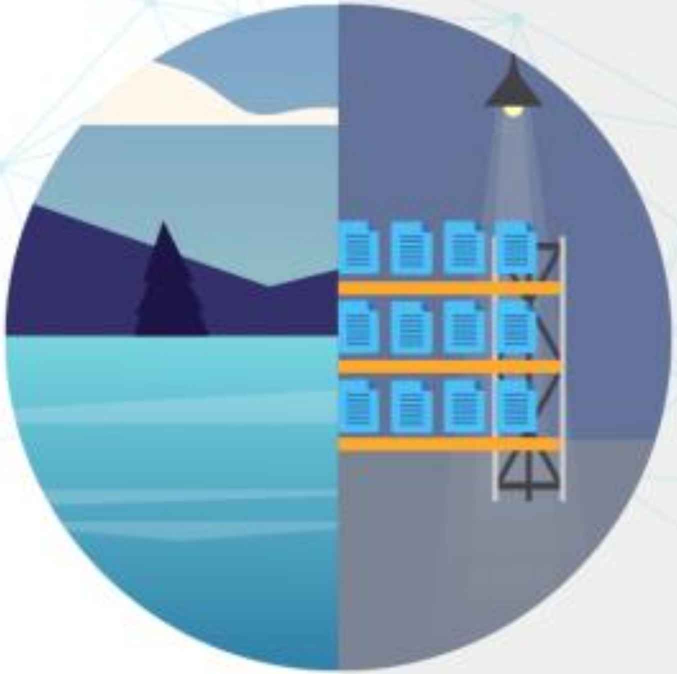
Data Lakes contain unstructured, semi structured and structured data with minimal processing. It can be used to contain unconventional data such as log and sensor data

Large

Data Lakes contain vast amounts of data in the order of petabytes. Since the data can be in any form or size, large amounts of unstructured data can be stored indefinitely and can be transformed when in use only

Undefined

Data in data lakes can be used for a wide variety of applications, such as Machine Learning, Streaming analytics, and AI



Refined

Data Warehouses contain highly structured data that is cleaned, pre-processed and refined. This data is stored for very specific use cases such as BI.

Smaller

Data Warehouses contain less data in the order of terabytes. In order to maintain data cleanliness and health of the warehouse, Data must be processed before ingestion and periodic purging of data is necessary

Relational

Data Warehouses contain historic and relational data, such as transaction systems, operations etc

Data Swamp

- A data lake turns into a data swamp when it becomes a disorganized, unmanaged dumping ground for data, rendering it unusable.
- This occurs due to a lack of governance, metadata, data quality controls, and ownership, resulting in chaotic, untrusted, and hard-to-find data that costs more than it provides.
- Consequences of a Data Swamp

Impossible Data Discovery: Users cannot find the data they need.

Loss of Trust: Analysts and data scientists cannot rely on the accuracy of the data.

Security & Compliance Risks: Poorly managed, unclassified data poses privacy risks.

High Costs, Low Value: Storage costs increase, while actionable insights decrease.

How to avoid data swamp in a data lake?

This slide depicts how to avoid a data swamp in a data lake by defining the checklist of questions to consider while establishing a data lake in the organization.

This slide is 100% editable. Adapt it to your needs and capture your audience's attention.

The Layers of a Data Lake

A well-designed data lake is organized into "zones" to prevent it from becoming a "Data Swamp" (a disorganized mess where data cannot be found).

1. **Bronze Layer (Raw Zone):** This is the landing area. Data is stored exactly as it comes from the source (logs, social media feeds, IoT sensors). No cleaning happens here.
2. **Silver Layer (Cleansed/Enriched Zone):** Data is filtered, cleaned, and standardized. For example, date formats are unified, and "null" values are handled.
3. **Gold Layer (Curated/Business Zone):** Data is organized into high-level business structures (like "Active Customers" or "Annual Revenue") ready for consumption by BI tools and executives.

Cloud storage

- Cloud storage is a method of storing data on remote servers that are accessed through the internet instead of saving data only on a local computer or physical device.
- These servers are maintained by cloud service providers, so users do not need to worry about hardware, maintenance, or data loss due to system failure.
- Cloud storage is highly scalable, which means storage space can be increased or decreased as per requirement. It also provides data backup and recovery features, helping protect data from accidental deletion or hardware damage.
- Security measures such as passwords, encryption, and access control are commonly used to keep data safe.
- Common examples of cloud storage services include Google Drive, OneDrive, Dropbox, and Amazon S3.

Data Processing

- Data processing is the stage in the data life cycle where raw data is cleaned, organized, and transformed into meaningful information for analysis. During data processing, we usually do things like:

  - Removing duplicate records
  - Correcting wrong values
  - Handling missing data
  - Converting text to numbers
  - Formatting dates and units
  - Sorting and grouping

Why Data Processing is Important?

Without data processing:

- Analysis gives wrong results
- Charts become misleading
- Decisions become risky

With data processing:

- Accurate insights
- Reliable reports
- Better decision-making

Batch Processing

- **Batch processing** is a method by which a computer system executes **large groups of repetitive, high-volume data tasks together (in batches)** rather than handling them one by one.
- A system **collects and stores data over a period of time**,
- Then it **processes all of it at once** during a scheduled window.
- **Example:-**
  An e-commerce site might receive orders throughout the day. Instead of processing each order instantly, the system groups all orders and handles *all of them together at night*-creating a single, efficient batch job.

Why Do We Use Batch Processing?

a. Minimizes Manual Effort

It automates repetitive tasks without constant human oversight.

b. Efficient Use of Resources

It schedules heavy computations during times when system resources are more available (like during off-peak hours).

c. Handles Large Volumes of Data

It's well suited for millions of records that would be inefficient to process individually in real time.

- ***You set it up once; it runs periodically with minimal supervision***

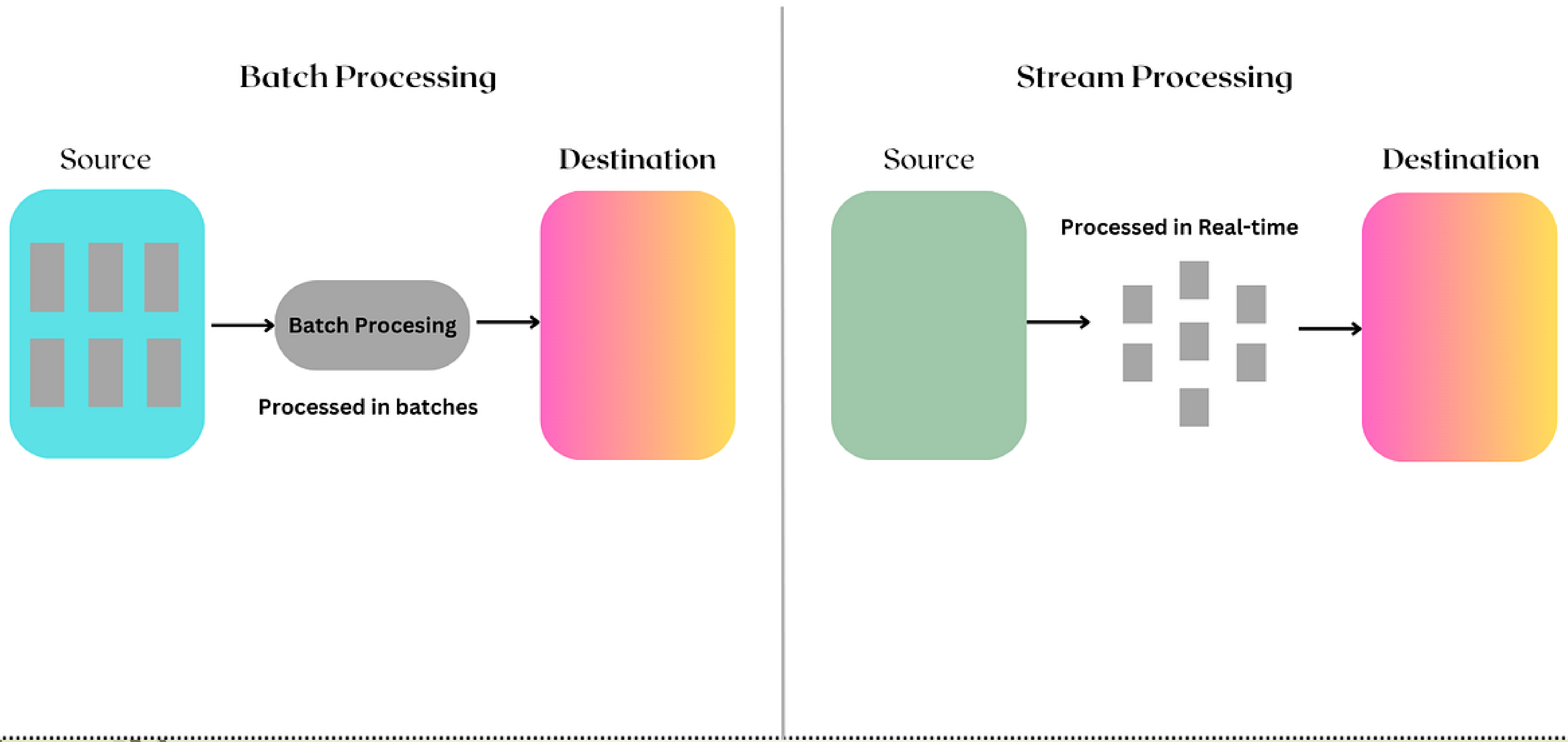
Stream processing:-

Stream processing is the processing of event data in real-time or near real-time.

The goal is to continuously take inputs of data (or “event”) streams, and immediately process (or transform) the data into streams of outputs.

The input is typically one or more event streams of data “in motion” containing information such as social media activity, emails, IoT sensor data from physical assets (e.g. self-driving cars), insurance claims, customer orders, bank deposits/withdrawals, and even emails. The types of streaming data are endless.

Batch vs Stream Processing



Web Scraping

- **Web scraping** is the process of **automatically visiting websites and collecting information from the web pages** the same information a human can see, but done by a program instead of a person.
- Imagine you open a website, scroll, copy prices, names, or tables, and paste them into Excel.
  Web scraping does **exactly this**, but faster and for thousands of pages.
- Scraping:
  - Product prices from Amazon
  - Job listings from job portals
  - News headlines from news websites
  - Exam notifications from university sites

When web scraping is useful

- Website **does not provide an API**
- Data is **publicly visible**
- You need **large-scale data collection**

Limitations

- Websites may block scrapers
- Layout changes can break the scraper
- Legal and ethical restrictions may apply

Web scraping is powerful but should be used **carefully and responsibly**.

Data Analysis

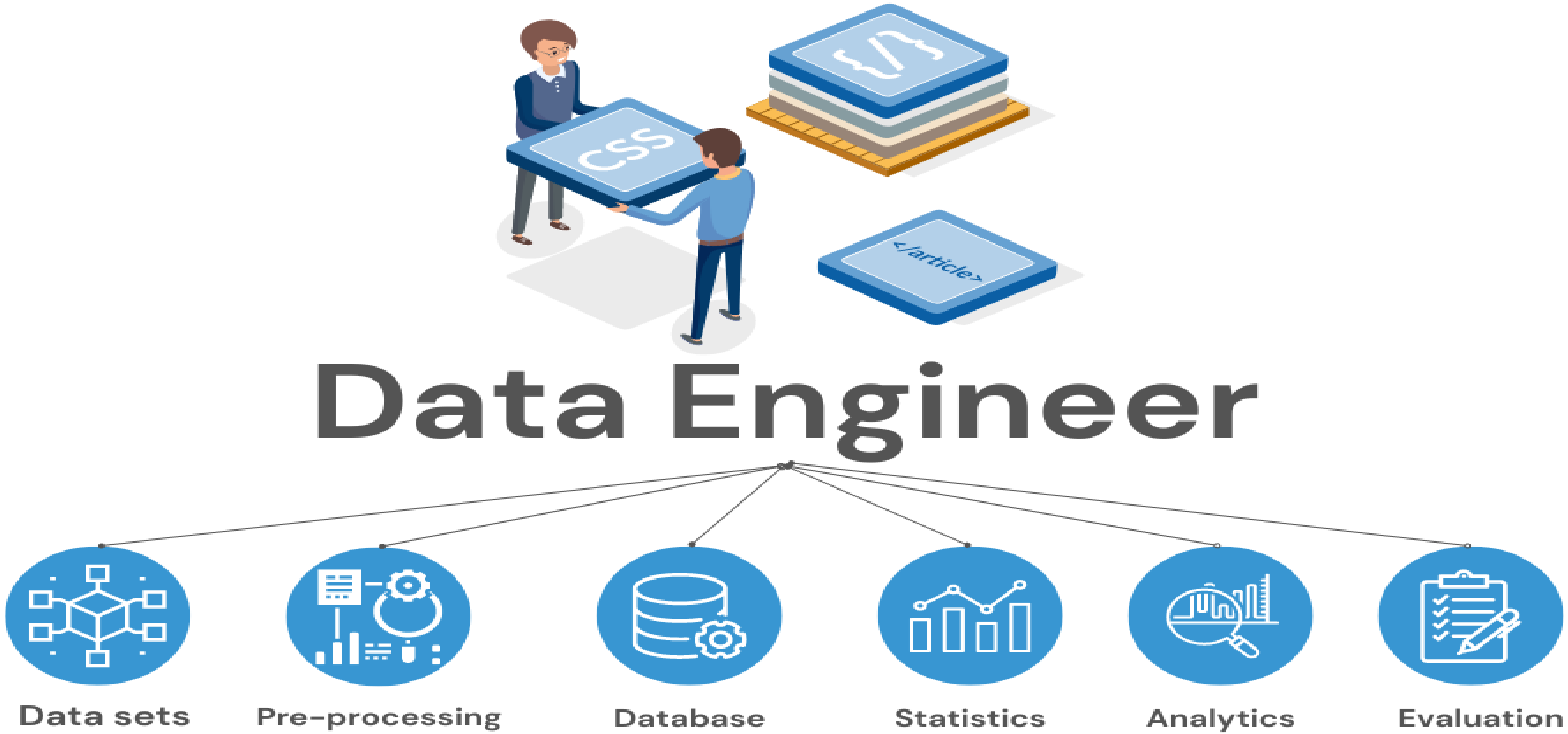
- Data analysis is the process of examining, organizing, and interpreting data to discover useful information, draw conclusions, and support decision-making.
- After data is collected and processed, analysis helps convert raw numbers into meaningful insights. It answers important questions such as *what happened, why it happened, and what can happen next*. Without data analysis, collected data remains just a set of values with no practical use.
- At this stage, the cleaned and processed data is examined to extract meaningful information. Data analysis focuses on understanding *what the data is saying*. This includes calculating totals, averages, percentages, comparisons, and identifying trends or patterns.
- For example, after collecting student marks data, analysis may involve finding the average marks, highest and lowest scores, or pass–fail percentages. The main goal of data analysis is to convert raw data into insights that can help answer questions and support decisions.
- Once data analysis is completed, the **next step in the data life cycle is data visualization.**

Data Visualization

- Data visualization is done to present the results of data analysis in a visual form such as charts, graphs, dashboards, or tables. While analysis gives insights in numbers, visualization makes those insights easy to understand at a glance. For instance, instead of showing average marks as numbers, a bar chart can visually compare performance across subjects or students, making patterns and differences clearer.
- In simple terms, **data analysis finds the answers**, and **data visualization shows those answers clearly**. That is why visualization always comes after analysis in the data life cycle because visuals are created based on analyzed data, not raw data. Together, these two steps help in better understanding, communication, and data-driven decision-making.

Data engineer

- “A **Data Engineer** is the professional responsible for designing, building, and maintaining the infrastructure that allows for the large-scale collection, storage, and analysis of data.”


CSS
</article>
Data Engineer
Data sets
Pre-processing
Database
Statistics
Analytics
Evaluation

Role of data engineer

- **Data Collection**
  A data engineer collects data from multiple sources such as databases, APIs, web applications, sensors, and logs. They make sure data is gathered automatically and regularly so that organizations always have updated information to work with.
- **Building Data Pipelines (ETL)**
  They design and maintain data pipelines that move data from source systems to storage systems. These pipelines ensure smooth data flow and handle large volumes of data efficiently without data loss.
- **Data Cleaning and Transformation**
  Raw data is often messy and unusable. Data engineers clean, filter, and transform data into a structured and meaningful format so that analysts and visualization tools can easily use it.
- **Data Storage Management**
  They decide how and where data should be stored, such as in relational databases, NoSQL databases, data lakes, or data warehouses. Proper storage design helps in fast querying and efficient data access.

- **Ensuring Data Quality and Reliability**
  Data engineers implement checks to ensure data accuracy, consistency, and completeness. This prevents incorrect or duplicate data from affecting analysis and decision-making.
- **Performance Optimization**
  They optimize databases and queries to improve system performance. Faster data access means quicker reports, dashboards, and visualizations.
- **Data Security and Access Control**
  Data engineers ensure that sensitive data is protected by applying security rules, encryption, and access permissions so that only authorized users can access the data.
- **Supporting Data Analysts and Scientists**
  They work closely with data analysts and data scientists by providing clean, well-organized data. This allows analysts to focus on insights and visualization instead of data preparation.

In short, a data engineer creates a strong and reliable data foundation that makes data analysis and data visualization possible.

Introduction to excel

- Microsoft Excel, a versatile spreadsheet tool from the Microsoft Office suite that organizes data into rows and columns. Whether we’re managing budgets, creating charts, or analyzing datasets,
- Excel helps us handle tasks efficiently. For example, we can list project tasks, calculate totals, and visualize data all in one file.
- Excel simplifies data tasks, from basic lists to complex analysis. Widely used in workplaces, it supports reports, project tracking and decision-making.

Core Excel Features

- Excel offers tools to manage data effectively:
- **Data Organization:** Store text, numbers or dates in cells. For example, list project deadlines in column A and statuses in column B.
- **Calculations:** Use formulas like =SUM(A1:A5) to add numbers or =A1*2 to multiply.
- **Formatting:** Enhance readability with fonts, colors or cell borders.
- **Charts:** Create visuals like bar or pie charts to summarize data.
- **File Management:** Save files as .xlsx and share them easily.

Role of excel in data visualization

- **1. Easy Conversion of Data into Visual Form**
  Excel allows users to quickly convert raw numerical data into visual formats such as charts and graphs. With just a few clicks, large tables of data can be transformed into visuals that are easier to understand and interpret, especially for beginners.
- **2. Wide Variety of Chart Options**
  Excel provides multiple chart types like column, bar, line, pie, and area charts. This helps users choose the most suitable chart based on the type of data and the insight they want to present, making comparisons and trends clearer.
- **3. Powerful Data Summarization Tools**
  Using features like PivotTables and Pivot Charts, Excel can summarize large datasets efficiently. These tools help in grouping, aggregating, and analyzing data before visualizing it, which improves accuracy and clarity.
- **4. Highlighting Key Insights**
  Conditional formatting and sparklines in Excel help highlight important data points such as high values, low values, or trends. This draws attention to critical information and supports faster decision-making.
- **5. Interactive and User-Friendly Visualization**
  Excel supports filters, slicers, and dynamic charts, making visualizations interactive. Users can explore data from different angles without technical skills, which makes Excel an effective tool for data-driven communication.

Cleaning data in excel

1. Handling missing values
2. Removing duplicate values
3. Convert Text to Numbers

Some functions in excel

- **SUM()** – Total marks
- AVERAGE() – Average marks
- MAX() / MIN() – Highest & lowest marks
- **COUNT()** only counts cells containing **numbers, dates, or times** (numeric data).
- **COUNTA()** (Count All) counts all **non-blank cells**, including text, numbers, errors, and logical values.
- IF() –logic
- IFERROR() – Handle blank cells
- ROUND()

Combo Charts

- While dealing with hierarchical data we need to combine two or more various chart types into a single chart for better visualization and analysis. These are known as "Combination charts" in Excel.

You can use a combination chart on the basis of the following points:

- Compare different data types that have different scales.
- Highlight trends between two datasets.
- Visualize targets and actuals for a better understanding of performance.

How to create a Combination Chart in Excel?

- Follow the below steps to create a combination chart in Excel:
- **Step 1**: Select the range of dataset you want in your dataset.
- **Step 2:** Go to the Insert tab in the Excel Ribbon.
- **Step 3:** Click on the "Recommended charts", and choose the combination of charts according to your need.
- **Step 4:** Customize the Chart as needed by right-clicking on various elements, such as data series, axis and labels.

Pivot Chart

- A Pivot Chart is a dynamic chart created from a Pivot Table. It helps us summarize, analyze, and visualize large data easily.
- In simple words:

Pivot Table = data summary
Pivot Chart = visual representation of that summary

- **Why Pivot Charts are Used**

Pivot Charts are useful because they:

  - Automatically update when data changes
  - Allow quick comparison of values
  - Work well with large datasets
  - Can be filtered easily using slicers

Slicer

- A **Slicer** is a visual filtering tool in **Microsoft Excel** that allows users to filter data in **Pivot Tables and Pivot Charts** easily using buttons instead of dropdown filters. It provides an interactive way to analyze data.

Why Use a Slicer?

- Makes filtering easier and more visual
- Shows current filter selection clearly
- Useful for dashboards
- Can connect multiple Pivot Tables

Normalization

- Normalization is the process of scaling numerical data to a common range or scale so that values can be fairly compared, analyzed, and visualized.
- In simple words, normalization changes the scale of data, not the meaning of the data.
- Following methods can be used to normalize the data:

a) Min-max normalization by setting new_min = 0 and new_max = 1.

b) z-score normalization using standard deviation.

c) z-score normalization using the mean absolute deviation

d) Normalization by Decimal scaling.

Denormalization

Denormalization is a database optimization technique that intentionally adds redundant data or groups data to improve read performance by reducing complex table joins. Applied after normalization, it balances data integrity with faster query speed. It is commonly used in read-intensive applications to improve response times.

Denormalization in data analytics is the strategic, intentional introduction of data redundancy into a database by merging tables to optimize read performance and simplify queries. It improves analytical speed for OLAP systems and dashboards, bypassing costly joins, but at the expense of increased storage needs and higher maintenance for updates.

a) Min–Max Normalization (0 to 1)

- Min–Max normalization **rescales data into a fixed range**, here **0 to 1**. The smallest value becomes **0**, the largest becomes **1**, and all other values fall in between.
- **Formula**
- $X' = \frac{X - X_{min}}{X_{max} - X_{min}}$
- **Step 1: Find minimum and maximum**
- **Step 2: Find values using Formula**

Question1: Use these methods to normalize the following group of data: 100, 500, 900, 1000, 12000.

a) min-max normalization by setting new_min = 0 and new_max = 1.

b) z-score normalization.

c) z-score normalization using the mean absolute deviation instead of standard deviation.

d) Normalization by decimal scaling.

MIN MAX- Solution

Step 1

Xmin = 100

Xmax = 12000

Step 2

Original Value	Normalized Value
100	(100−100)/(12000−100) = **0**
500	400/11900 = **0.0336**
900	800/11900 = **0.0672**
1000	900/11900 = **0.0756**
12000	11900/11900 = **1**

b) Z-Score Normalization (Standard Deviation)

- Z-score normalization shows **how far a value is from the mean**, measured in **standard deviations**.
- **Formula**

$$Z = \frac{X - \mu}{\sigma}$$

- **Step 1: Calculate Mean (μ)**

$$\mu = \frac{\sum_{i=1}^{N} X_I}{N}$$

- **Step 2: Calculate Standard Deviation (σ)**

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

- **Step 3: Calculate z scores**

$$Z = \frac{X - \mu}{\sigma}$$

- **Step 1: Calculate Mean (μ)**

$$mean\ (\mu) = \frac{100 + 500 + 900 + 1000 + 12000}{5} = \frac{14500}{5} = 2900$$

- **Step 2: Calculate Standard Deviation (σ)**

Value	X − μ	(X − μ)²
100	-2800	7,840,000
500	-2400	5,760,000
900	-2000	4,000,000
1000	-1900	3,610,000
12000	9100	82,810,000

$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}} = \sqrt{\frac{104020000}{5}}$$

$\sigma \approx 4561$

Step 3: Calculate z Scores

$$Z = \frac{X - \mu}{\sigma}$$

Value	Z-Score
100	-0.61
500	-0.53
900	-0.44
1000	-0.42
12000	1.99

c) Z-Score Normalization using Mean Absolute Deviation (MAD)

- Instead of standard deviation, this method uses **average absolute distance from the mean**. It is **less sensitive to outliers**.
- **Step 1: Calculate Mean (μ)**

$$\mu = \frac{\sum_{i=1}^{N} X_I}{N}$$

- **Step 2: Calculate Mean Absolute Deviation (MAD)**

$$\text{MAD} = \frac{\sum |X - \mu|}{N}$$

- **Step 3: Calculate normalized values**

$$Z = \frac{X - \mu}{MAD}$$

c) Z-Score Normalization using Mean Absolute Deviation (MAD)

Step 1: Calculate Mean (μ)

$$mean\ (\mu) = \frac{100 + 500 + 900 + 1000 + 12000}{5} = \frac{14500}{5} = 2900$$

- **Step 2: Calculate MAD**

Value	$\lvert X - \mu \rvert$
100	2800
500	2400
900	2000
1000	1900
12000	9100

$$\text{MAD} = \frac{\Sigma \lvert X - \mu \rvert}{N} = 18200/5$$

MAD=3640

Step 3: Calculate normalized values

Value	Normalized
100	-0.77
500	-0.66
900	-0.55
1000	-0.52
12000	2.50

Step 3: Calculate normalized values

$$Z = \frac{X - \mu}{MAD}$$

Normalization by Decimal Scaling

- Decimal scaling **moves the decimal point** until all values lie between **−1 and 1**.
- **Formula**

$$X' = \frac{X}{10^j}$$

- Where **j** is the number of digits in the largest value.
- **Step 1: Find largest value**
- **Step 2: Count the number of digits**
- **Step 3: Find the normalized value**

Solution

- Largest digit

12000 → 5 digits → j = 5

Apply formula

$$X' = \frac{X}{10^j}$$

Value	Normalized
100	0.001
500	0.005
900	0.009
1000	0.01
12000	0.12

Use the following methods to normalize the given data set:

Data:

50, 150, 300, 450, 2000

Perform:

a) Min–Max normalization by setting **new_min = 0** and **new_max = 1**

b) Z-score normalization

c) Z-score normalization using **Mean Absolute Deviation (MAD)**

d) Normalization by **Decimal Scaling**

Mean, median, mode for ungrouped data

- The following marks (out of 100) were obtained by 8 students in a test:

45, 78, 56, 89, 45, 67, 72, 56

a) Find the **Mean**
b) Find the **Median**
c) Find the **Mode**

Solution

45, 78, 56, 89, 45, 67, 72, 56

- **Mean** = 508 / 8 = **63.5**
- **Median**

Arrange in ascending order **:** 45, 45, 56, 56, 67, 72, 78, 89

Median (for even no of elements)= $\dfrac{\left(\frac{n}{2}{}^{th}\ value + \left(\frac{n}{2}+1\right)^{th}\ value\right)}{2}$

$\frac{N}{2} = \frac{8}{2} = 4 \Rightarrow \frac{(4^{\text{th}}\ \text{term}+5^{\text{th}})}{2}$

- (56+67)/2=123/2=**61.5**
- **Mode:** Most repeated value

45 and 56

Mean, median, mode for grouped data

The following table shows marks obtained by students:

Find:

1. Mean
2. Median
3. Mode

Marks (Class Interval)	Frequency (f)
0 – 10	5
10 – 20	8
20 – 30	12
30 – 40	15
40 – 50	10

- **Step 1: Mean for Grouped Data**
- **Formula:** Mean $= \frac{\sum fx}{\sum f}$

Where: f = frequency, x = class midpoint

Step 2: Find mid points (x)

$$x = \frac{upper\ bound + lower\ bound}{2}$$

Class	f	Midpoint (x)	f × x
0–10	5	5	25
10–20	8	15	120
20–30	12	25	300
30–40	15	35	525
40–50	10	45	450
	$\sum f = 50$		$\sum f\,x = 1420$

Mean $= \frac{\sum fx}{\sum f} = 1420/50 =$ **28.4**

Median for Grouped Data

Formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$$

Where:

L = Lower limit of median class

N = Total frequency

cf = Cumulative frequency before median class

f = Frequency of median class

h = Class width

- **Find Cumulative Frequency (CF)**
- **Find N/2**

N = 50
N/2 = 25

- The class where CF just exceeds 25 is:

30–40, So this is the **Median Class.**

- **Now the values are:**

L = 30
cf = 25
f = 15
h = 10

- Apply formula : Median $= L + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$
- 30+((25−25)/15)×10
- Median = 30 + 0×10
  Median = **30**

Class	f	CF
0–10	5	5
10–20	8	13
20–30	12	25
30–40	15	40
40–50	10	50
	N=50	

Mode for Grouped Data

- **Formula:**

$$Mode = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

- Where:

L = Lower limit of modal class

f_1 = Frequency of modal class

f_0 = Frequency of previous class

f_2 = Frequency of next class

h = Class width

- **Find Modal Class**

Highest frequency = 15

So modal class = **30–40**

- **Values:**
- $L = 30$
  $f_1 = 15$
  $f_0 = 12$
  $f_2 = 10$
  $h = 10$
- Apply Formula: $Mode = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$
- Mode= $30 + \left(\frac{15-12}{2(15)-12-10}\right) \times 10$

$$= 30 + \left(\frac{3}{30 - 22}\right) \times 10$$

$$= 30 + \left(\frac{3}{8}\right) \times 10$$

$$= 30 + 3.75$$

Mode = 33.75

Class	f
0–10	5
10–20	8
20–30	12
30–40	15
40–50	10
	N=50

Parameters	OLTP	OLAP
Purpose	It is a system for processing large volumes of real-time transactional data.	It is a system for the multidimensional analysis of consolidated business data.
Usage	It is used for adding, deleting, or updating databases to keep the data up-to-date.	It is used to make business decisions through queries and complex analyses of large amounts of data.
Focus	The system is more focused on transactional data maintenance and less on data analysis.	The system is focused on data analysis and not on maintaining day-to-day transactions.
Data Source	OLTP sources data from traditional database management systems.	OLAP has multiple data sources, which include real-time and historical databases, including OLTP.
Data Type	The data consists of a large number of short transactions.	The system processes large volumes of data from multiple sources.
Processing Time	Very low processing time at the scale of a few milliseconds.	Depending on the query, processing time is not as fast as OLTP systems and may range from a few seconds to hours.
Query	Related to adding, deleting, and updating data.	Related to data analysis.
Availability	OLTP systems are available round-the-clock and updated frequently to maintain data integrity.	OLAP systems don't need to be updated so frequently since their functions are analytic in nature.
Normalization	Data tables are normalized.	Data tables are not normalized.
Backup	Requires constant backup and recovery.	Can be backed up less frequently.
User volume	Supports large user volume simultaneously.	Accommodates multiple users but doesn't have a large user volume like OLTP.
Operations	Allows both read and write operations.	Usually supports read-only operations.
Process	Processes day-to-day data quickly.	Processes analytical queries consistently and at a fast pace.

Data processing and transformation

ETL Process (Extract–Transform–Load)

- ETL is a critical data integration process that retrieves raw data from various sources, converts it into a consistent format, and loads it into a centralized system like a data warehouse.
- It is essential for ensuring data quality, enabling accurate analysis, and driving informed business decisions. ETL is used in **Data Warehousing, OLAP, Business Intelligence, and Data Analytics** to prepare raw data for analysis.
- **ETL vs. ELT**

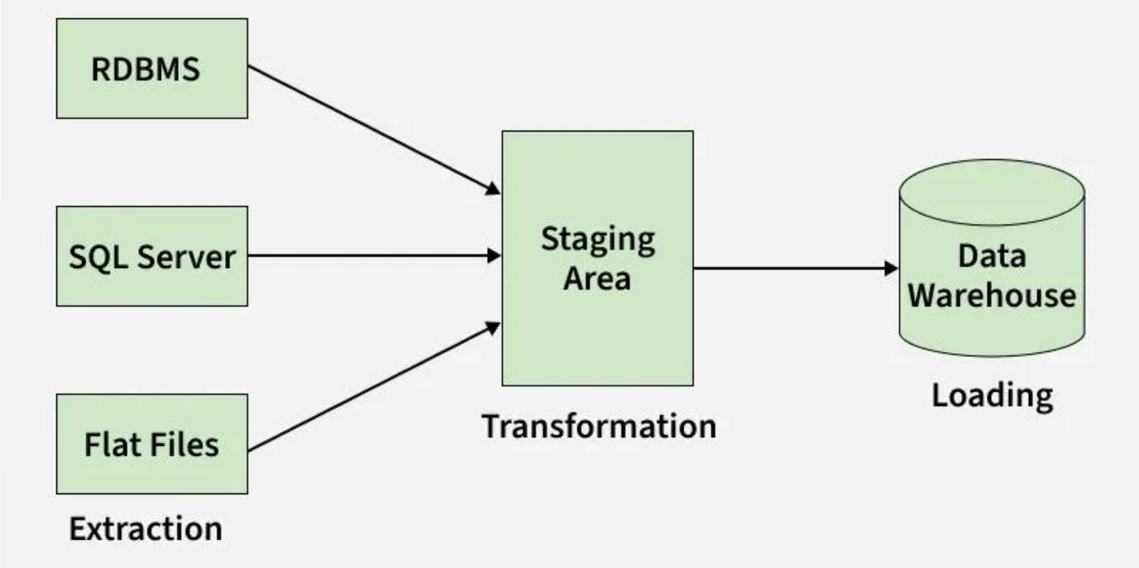
While ETL transforms data before loading,

ELT loads raw data directly into a modern

cloud data warehouse

first. ELT is often faster and more flexible

for large-scale, unstructured datasets.



1. Extraction

The Extract phase is the first step in the ETL process, where raw data is collected from various data sources. These sources can be diverse, ranging from structured sources like databases (SQL, NoSQL), to semi-structured data like JSON, XML or unstructured data such as emails or flat files. The main goal of extraction is to gather data without altering its format, enabling it to be further processed in the next stage.

2. Transformation

The Transform phase is where the magic happens. Data extracted in the previous phase is often raw and inconsistent. During transformation, the data is cleaned, aggregated and formatted according to business rules. This is a crucial step because it ensures that the data meets the quality standards required for accurate analysis.

- ***Common transformations include:***

Data Filtering: Removing irrelevant or incorrect data.

Data Sorting: Organizing data into a required order for easier analysis.

Data Aggregating: Summarizing data to provide meaningful insights (e.g., averaging sales data).

3. Loading

Once data has been cleaned and transformed, it is ready for the final step: Loading. This phase involves transferring the transformed data into a data warehouse, data lake or another target system for storage.

Depending on the use case, there are two types of loading methods:

- **Full Load:** All data is loaded into the target system, often used during the initial population of the warehouse.
- **Incremental Load:** Only new or updated data is loaded, making this method more efficient for ongoing data updates.
- **NOTE**

Pipelining in ETL Process

Pipelining in the ETL process involves processing data in overlapping stages to enhance efficiency. Instead of completing each step sequentially, data is extracted, transformed and loaded concurrently. As soon as data is extracted, it is transformed and while transformed data is being loaded into the warehouse, new data can continue being extracted and processed.

Importance in Data Processing and Pipelines

- **Data Quality and Consistency:** ETL cleanses data, removing duplicates and errors to ensure data accuracy.
- **Integration of Disparate Sources**: It consolidates, merges, and harmonizes data from multiple, incompatible systems into a unified view.
- **Scalability and Performance**: Pipelines can be automated and scheduled to handle large volumes of data, enhancing efficiency.
- **Regulatory Compliance**: It allows for data auditing and the removal of sensitive information (e.g., for GDPR/HIPAA) before storage.
- **Support for BI and Analytics**: By providing a structured, reliable data source, ETL enables efficient reporting and advanced machine learning.

Data modelling

- An **ER Diagram** is a graphical representation of the structure of a database.
  It shows how data is organized into entities, their attributes, and the relationships between them.
- It is mainly used during the **data modeling stage** before creating actual database tables.

Purpose of ER Diagram

  - To design database structure
  - To visualize data relationships
  - To reduce errors before implementation

- The Entity-Relationship Model (ER Model) is a conceptual model for designing a database. This model represents the logical structure of a database, including entities, their attributes, and relationships between them.
- **Entity:** An object that is stored as data. E.g: *Student*, *Course*, or *Company*.
- **Attribute:** Properties that describe an entity. E.g: *StudentID*, *CourseName*, or *EmployeeEmail*.
- **Relationship:** A connection between entities. E.g: *Student* enrolls in a *Course*.

Key Components of an ER Diagram

- Entity (Rectangle): Represents a real-world object, concept, or person about which data is stored (e.g., "Student," "Product").

  Strong Entity: Independent, uniquely identified by a primary key.

  Weak Entity: Depends on a strong entity for existence.

- Attribute (Oval): Represents properties or characteristics of an entity (e.g., Student Name, Student ID).

  Key Attribute: Uniquely identifies an entity instance (underlined).

  Multivalued Attribute: Can have multiple values (double oval).

  Derived Attribute: Calculated from other attributes (dashed oval).

- **Relationship (Diamond):** Illustrates how entities are connected or interact (e.g., a student "enrolls" in a course).
- **Cardinality (Lines/Numbers):** Defines the numerical relationship between entities (e.g., one-to-one, one-to-many, many-to-many).

Question :

- Use the following methods to normalize the given data set:Data: 50, 200, 450, 800, 5000

a) Min-max normalization with new_min = 0 and new_max = 1

b) Z-score normalization

c) Z-score normalization using Mean Absolute Deviation (MAD)

d) Normalization by decimal scaling

Creating dashboards using Pivot Tables and Slicers